

## PRACTICAL 1

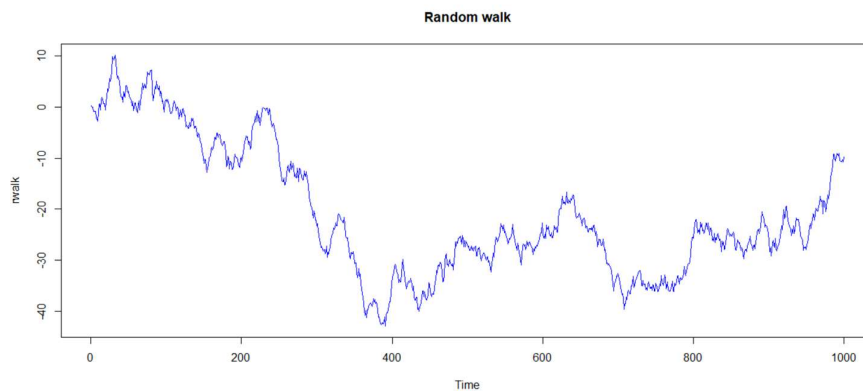
Simulate a random walk model. Also plot a time series & correlogram. Use a suitable test to check stationarity of generated time series.

### Code & Output

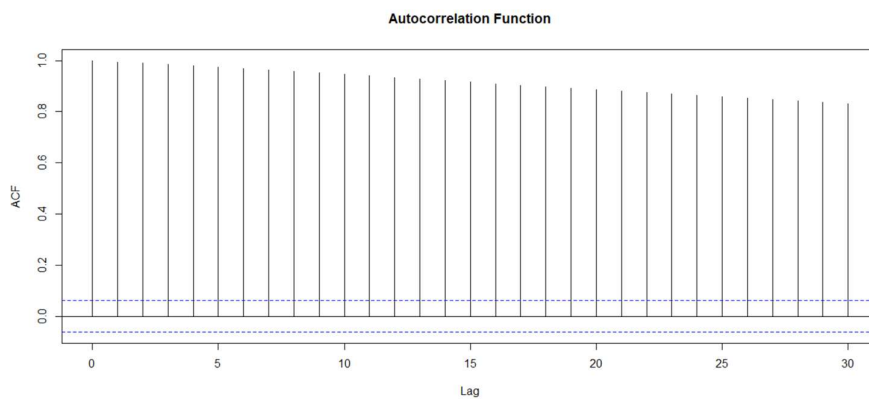
```
install.packages("tseries")  
library(tseries)
```

```
sample = rnorm(1000)  
rwalk = cumsum(sample)
```

```
ts.plot(rwalk, col = 'blue', main = 'Random walk')
```



```
acf(rwalk, main = "Autocorrelation Function")
```



```
adf.test(rwalk, alternative = "s")
```

Augmented Dickey-Fuller Test

data: rwalk

Dickey-Fuller = -1.0953, Lag order = 9, p-value = 0.9233

alternative hypothesis: stationary

## PRACTICAL 2

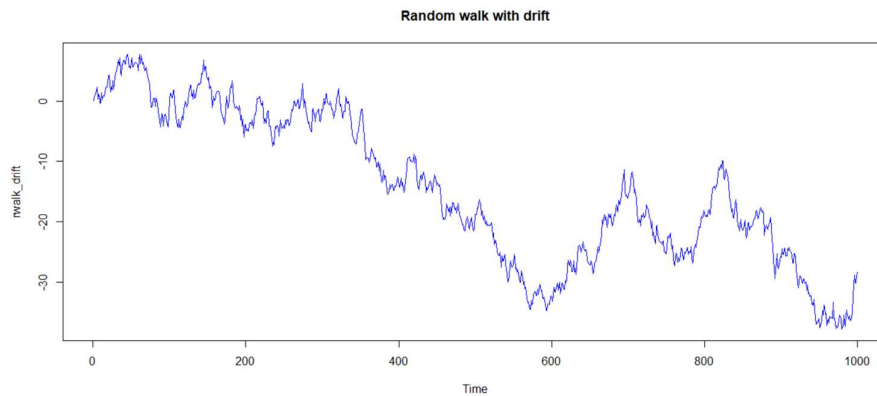
Simulate a random walk model with drift given by  $y_{t-1} + a + \varepsilon_t$  with  $a = 0.1$ . Check stationarity of generated time series.

### Code & Output

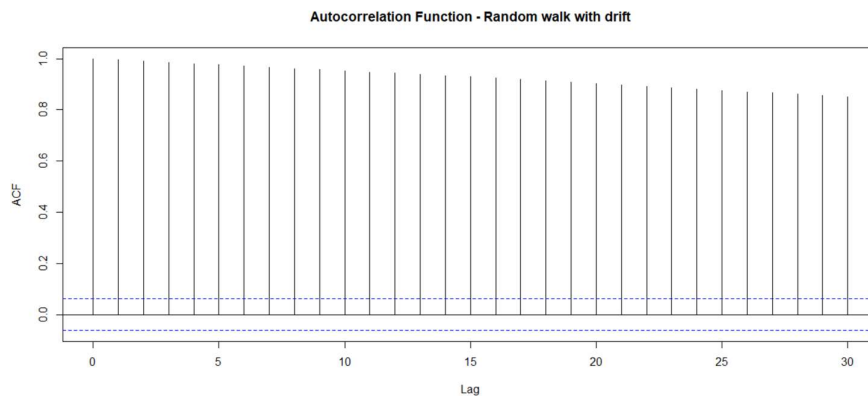
```
install.packages("tseries")  
library(tseries)
```

```
sample = rnorm(1000)  
rwalk_drift = cumsum(sample) + 0.1
```

```
ts.plot(rwalk_drift, col = 'blue', main = "Random walk with drift")
```



```
acf(rwalk_drift, main = "Autocorrelation Function - Random walk with drift")
```



```
adf.test(rwalk_drift, alternative = "s")
```

Augmented Dickey-Fuller Test

data: rwalk\_drift

Dickey-Fuller = -2.1889, Lag order = 9, p-value = 0.4984

alternative hypothesis: stationary